OSI MODEL

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INTRODUCTION TO THE OSI MODEL

- Definition: A conceptual framework developed by the ISO (International Organization for Standardization) that describes how data is transmitted over a network.
- Purpose: To standardize communication between different systems regardless of their underlying architecture.
- Structure: It consists of 7 distinct layers, each with specific functions and protocols.

THE UPPER LAYERS (APPLICATION, PRESENTATION, SESSION)

- These layers focus on user interaction and data formatting.
- Layer 7 - Application: The layer closest to the user. It provides network services to applications (e.g., browsers, email clients).
- Protocols: HTTP, FTP, SMTP, DNS.
- Layer 6 - Presentation: Handles data translation, encryption, and compression. It ensures that data sent from the application layer of one system can be read by the application layer of another.
- Key task: Formats data (e.g., JPEG, GIF, MPEG).
- Layer 5 - Session: Manages the "dialogue" between computers. It establishes, maintains, and terminates connections between applications.

THE HEART OF THE MODEL (TRANSPORT LAYER)

- Layer 4 - Transport: Responsible for end-to-end communication, flow control, and error correction.
- Data Unit: Segments.
- Key Protocols:
- TCP (Transmission Control Protocol): Connection-oriented and reliable.
- UDP (User Datagram Protocol): Connectionless and fast.

THE LOWER LAYERS (NETWORK, DATA LINK, PHYSICAL)

These layers handle the actual movement of data across the physical medium.

- Layer 3 - Network: Responsible for logical addressing and routing. It determines the best physical path for the data.
  - Device: Router.
  - Data Unit: Packets.
- Layer 2 - Data Link: Provides node-to-node data transfer and handles physical addressing (MAC addresses).
  - Device: Switch/Bridge.
  - Data Unit: Frames.
- Layer 1 - Physical: The actual hardware level. It deals with the transmission of raw bitstreams over a physical medium (cables, radio waves).
  - Data Unit: Bits.

DATA ENCAPSULATION PROCESS

This slide is great for showing how data travels down the stack.

1. Data starts at the Application layer.
2. It becomes Segments at the Transport layer.
3. It becomes Packets at the Network layer.
4. It becomes Frames at the Data Link layer.
5. It is sent as Bits at the Physical layer.

CONCLUSION

- The OSI model remains the most effective tool for troubleshooting network issues by isolating the problem to a specific layer.
- It provides a universal language for network engineers and developers.
- Even as technology evolves (to 5G or Cloud computing), the fundamental logic of the 7 layers remains the backbone of communication.

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THANK YOU